

Santiago Miranda

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Developer History ✓

2025: Open Source Development

Embedded Tutorial: [micrors/esp32-c6](#) is a book for Rust Embedded development for newcomers. The book is free and open source.

FSF: Contributor to the [Machine Learning position](#) that the Freedom of Software Foundation will publish in the near future.

Web Standards: [Request for position for WebNN](#) to Mozilla. Specification issues and feedback [WebNN repository](#).

Rust Programming: [mdbook-preprocessor for fetching remote markdown](#).

TNT: Typescript implementation of the paper "*TNT: A Solver for Large Dense Least-Squares Problems*". The code can be found at [santi-mir/fit-tnt](#).

2024: Research Associate at Glasgow University

Robotics and Artificial Intelligence, The [Digital Chemistry Group](#) is a research group led by [Leroy Cronin](#).

My contribution: extending a chemical Variational Auto-Encoder (CVAE), that encodes molecules into vectors. It was trained in a High Performance Computing Cluster. The code is private due to Glasgow University's licensing policies (I can show it in an interview). Reference project: at [aspuru-guzik-group/chemical_vae](#).

For the application to this job, I made a brief video-presentation that summarises most of my work. The presentation is [on Youtube](#).

2022-2024: Typescript Developer at Zakodium

Binary Parsers for Chemical Data. [NMRium](#) is a web-platform for NMR (Nuclear Magnetic Resonance) visualisation and analysis. My role involved writing binary data parsers for NMR data. For example, a Varian Converter to parse (Varian/Agilent) data.

Image analysis. Anonymisation of personal information in Identity Cards, for Interpol and other organisations. A [public playground](#) was used as proof-of-concept.

This project involved several interesting challenges:

1. Creating a database of IDs,
2. Augmenting and cleaning up the data,
3. Fine-tune neural networks to remove signatures, faces and barcodes,
4. Compress models to a minimum (700 KB) to reduce the latency and loading time (for example, using quantisation from floating point to integers.)
5. New classification networks in PyTorch (from scratch), to fix the perspective of the ID card automatically.

Optimisation. Fast-Combinatorial Non-Negative Least Squares, published in Zenodo and available at [mljs/fcnls](#).

2021-2022: Projects for Cheminfo and EPFL

Binary Parsers. [wdf-parser](#) converts Raman binary files to JSON-like objects. A similar project is [spc-parser](#).

IOBuffer. A project I am involved in is [IOBuffer](#), a Typescript library for manipulating binary data.

File Handling. To harmonise the way NodeJS and FileList Web API handle files we wrote [filelist-utils](#) (with many collaborators).

Smaller Projects. Simple recursive directory reading and side projects published to theNPM (Node Package Registry).



2020-2022: Developer Hubs

Besides my work for EPFL and Zakodium, I remained active in programming forums throughout 2020-2024.

For example, I helped others in the [MongoDB Developers Forum](#).

2017-2019: Chemistry Teacher

Introduction to Chemistry at *Instituto Superior de Formación Docente* (Institute for Teacher Training), Argentina. The role consisted of teaching chemistry to aspiring High School teachers. It involved planning and supervising laboratory experiments, delivering lectures, and grading the students.

Higher Education ✓

2015-2017

MSc in Physical Chemistry at University of La Plata, Argentina.

Thesis: *Thermodynamics of chemical reactions in condensed phase*. The thesis involved integrating packages for conformer generation, solvent-effect simulation, and quantum mechanics and statistics. Programming Language: Python.

The thesis was developed at the Centre For Inorganic Chemistry and supervised by Martin Lavecchia (lavecchia@gmail.com) and Carlos Franca.

2012-2015

Chemistry Degree at University of La Plata, Argentina.

The first 3 years were in general chemistry with a strong mathematical background (physics, calculus, linear algebra.)

Hobbies ✓

Gap Year

During part of 2020 (lockdown) I volunteered at remote places like farms. In my spare time, I learnt Javascript. As a learning exercise during this process, I [coded a blog](#). Later on, this experience helped me land my first job as a Typescript Developer.

Time in Nature

I enjoy jogging or walking outdoors.

To start the day, I do meditation and breathing exercises, sometimes at home, and sometimes in Nature, depending on the weather.

Writing

During 2025 I learnt Rust and experimented with ESP32s, and wrote a brief book for beginners about how to set microcontrollers up for programming with Rust (on Unix systems).

During 2026, I [started a blog](#) on Explainable AI and AI for chemistry.